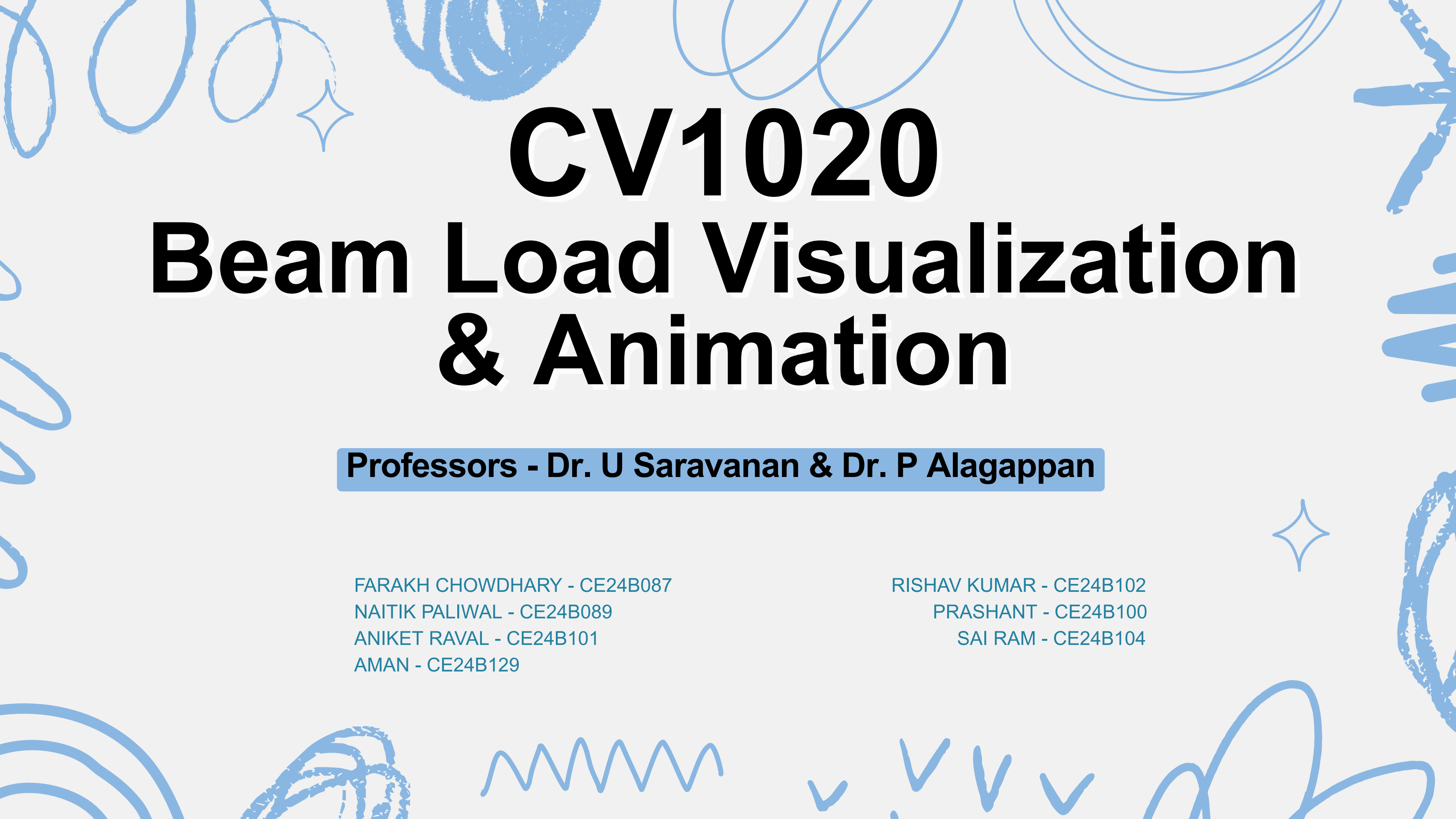




CV1020

Beam Load Visualization & Animation

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Project vision and mission

- To develop a comprehensive analytical and computational framework using MATLAB for evaluating the response of beams under various loading conditions.
 - This includes the generation of Shear Force Diagrams (SFD), Bending Moment Diagrams (BMD), and beam deflection.
 - The project aims to enhance understanding, improve accuracy, and analyzing process in civil engineering through automation and visualization.
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PROBLEM

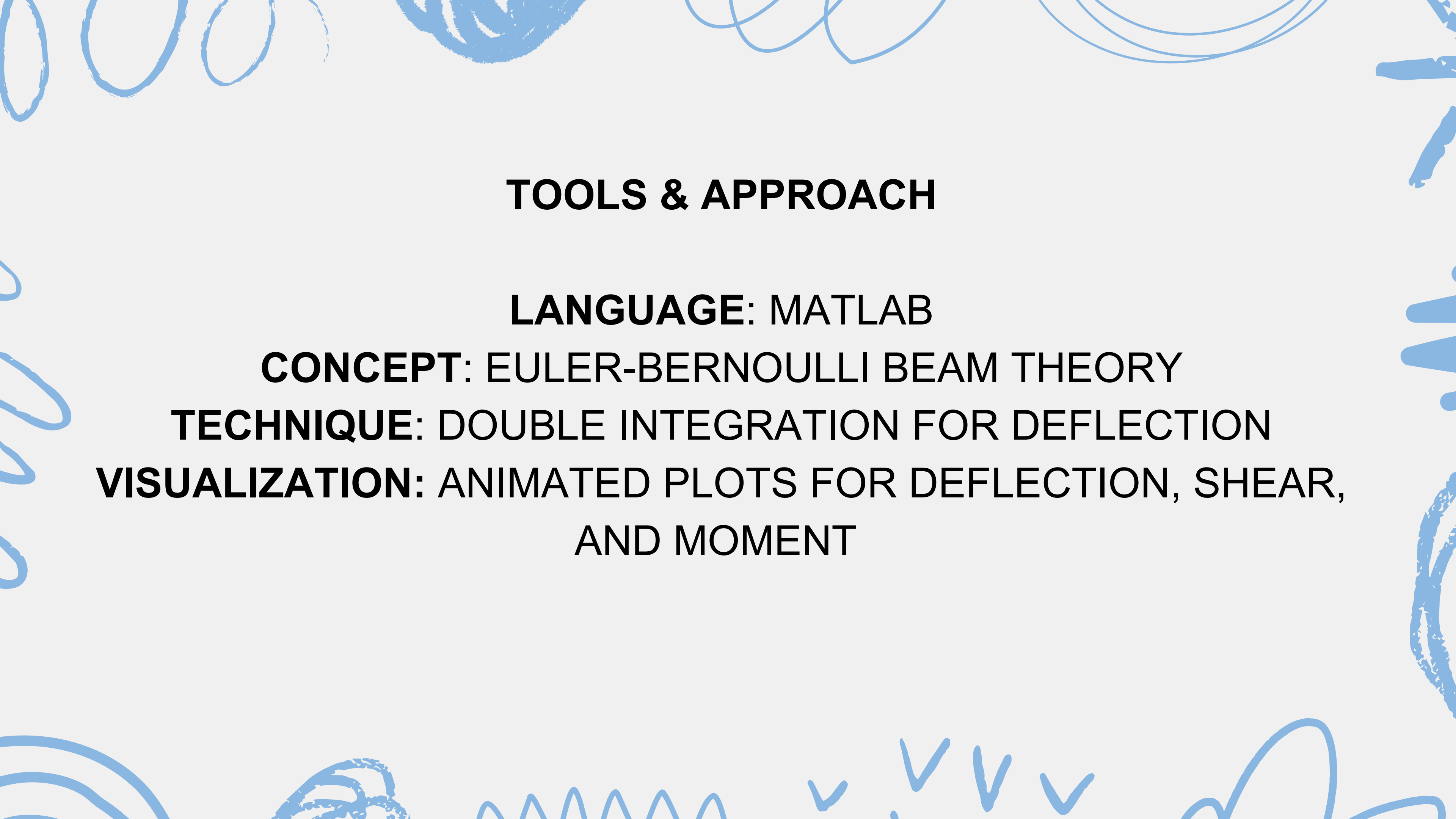
Traditional methods for analyzing beam behavior are static, abstract, and often difficult to understand—especially when dealing with complex loading and boundary conditions. This makes it hard for learners and even practitioners to visualize how internal forces and deflection change along the beam.

NEED

There is a need for an intuitive, interactive, and visual approach to beam analysis that helps users understand shear force, bending moment, and deflection in real-time, improving learning and aiding in structural design decisions.

GOAL

To develop a MATLAB-based simulation tool that dynamically computes and visualizes Shear Force Diagrams (SFD), Bending Moment Diagrams (BMD), and deflection curves for beams under various loading and support conditions.



TOOLS & APPROACH

LANGUAGE: MATLAB

CONCEPT: EULER-BERNOULLI BEAM THEORY

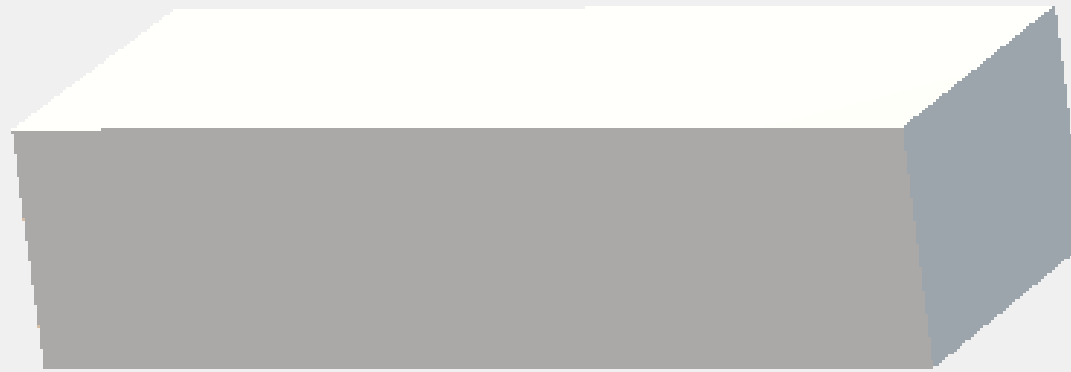
TECHNIQUE: DOUBLE INTEGRATION FOR DEFLECTION

VISUALIZATION: ANIMATED PLOTS FOR DEFLECTION, SHEAR,
AND MOMENT

DESCRIPTION

Formula Used:

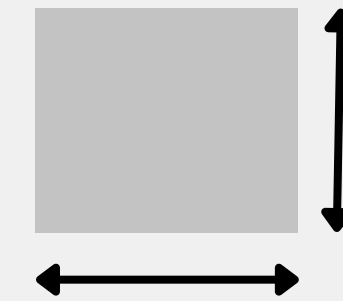
$$EI \frac{d^2 y(x)}{dx^2} = M(x)$$



Material Property:

Material : Steel

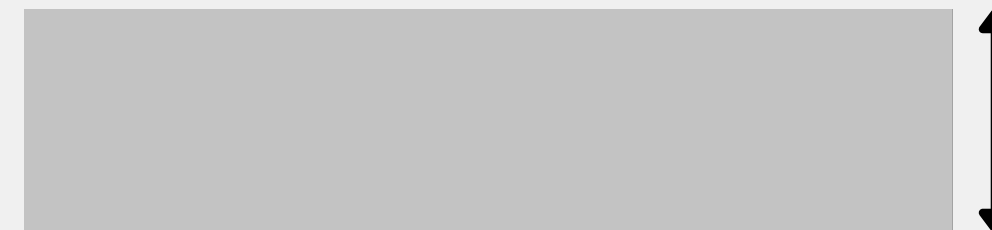
E : $200 \times 10^9 \text{ Pa}$



Height = 0.084m



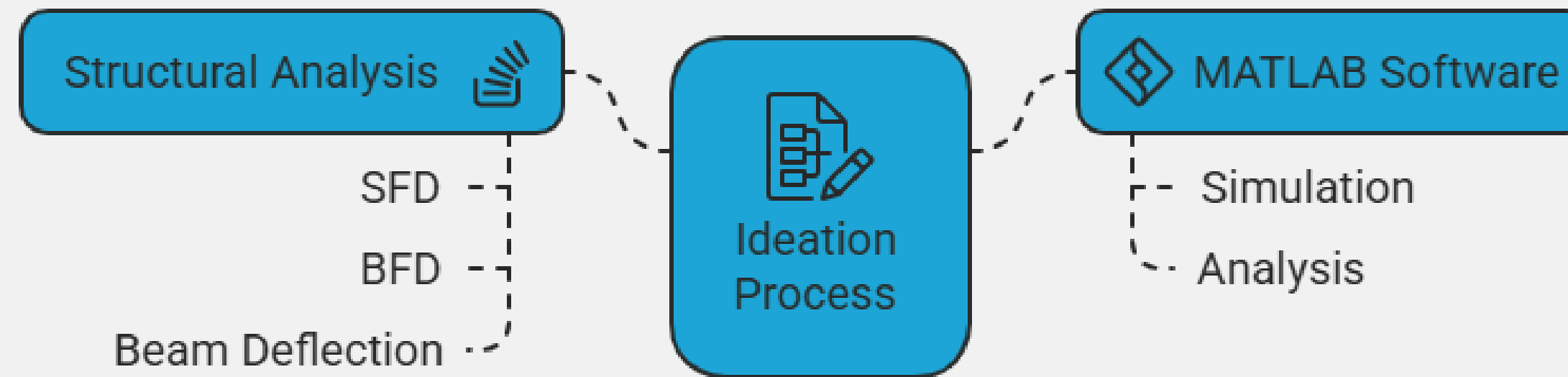
width (w) = 0.1m



Length (l) = 10m

Height (h) = 0.084m

Ideation Process for Beam Deflection Project



01

Problem Identification & Research

- Met with professor to understand project goals.
- Identified the need to help students visualize beam behavior under various loads.
- Researched beam types, load cases, and existing visualization tools.

02

Idea Generation & Selection

- Brainstormed multiple solutions: static diagrams, 3D models, interactive tools.
- Evaluated ideas based on feasibility, educational value, and usability.
- Chose to develop an interactive animation showing beam deflection and internal

03

Design & Development

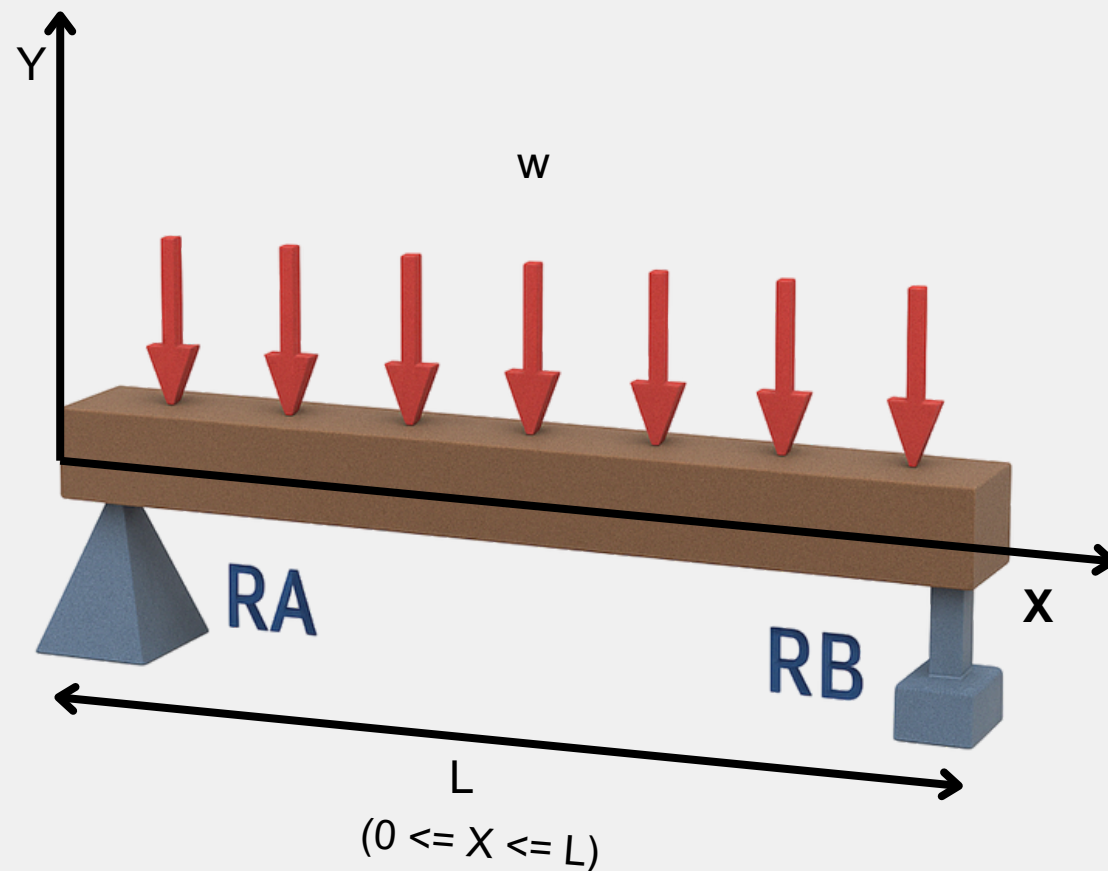
- Sketched user interface and planned software tools (e.g., Python, Blender).
- Built models and animations for different loading conditions.
- Added features like sliders for load adjustment and real-time visualization.

04

Testing, Feedback & Finalization

- Tested the tool with peers and gathered feedback.
- Refined the design and fixed usability issues.
- Finalized the project and presented it to the professor/class.

Uniform Distributed Load



Equations

Reaction Forces
 $R(A) = R(B) = \omega L/2$

Shear Force:
 $V(x) = \omega L/2 - \omega x$

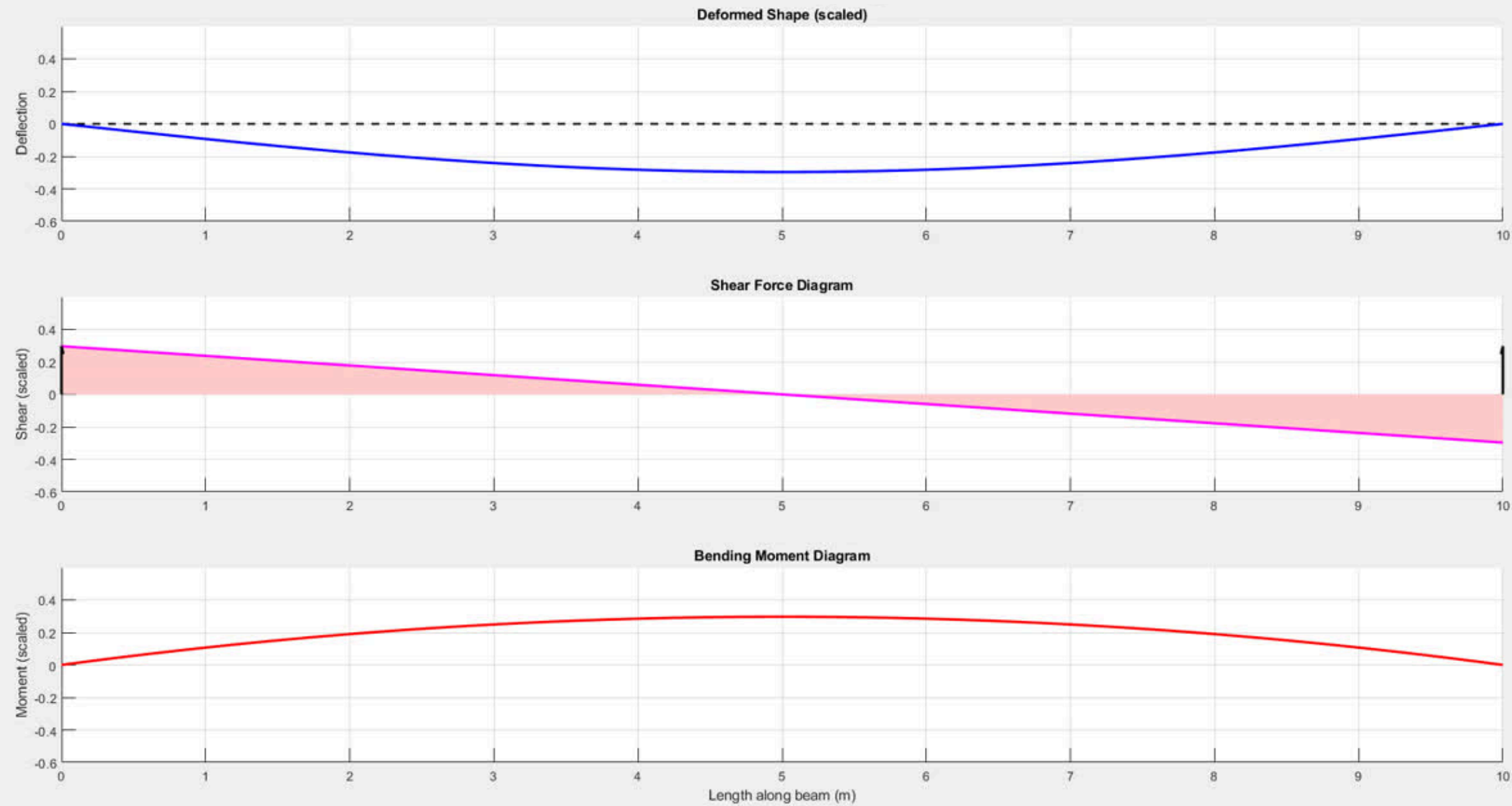
Bending Moment:
 $M(x) = \omega x(L - x)/2$

Deflection:
 $\delta y(x) = \omega x (-L^3 + 2Lx^2 - x^3)/24EI$

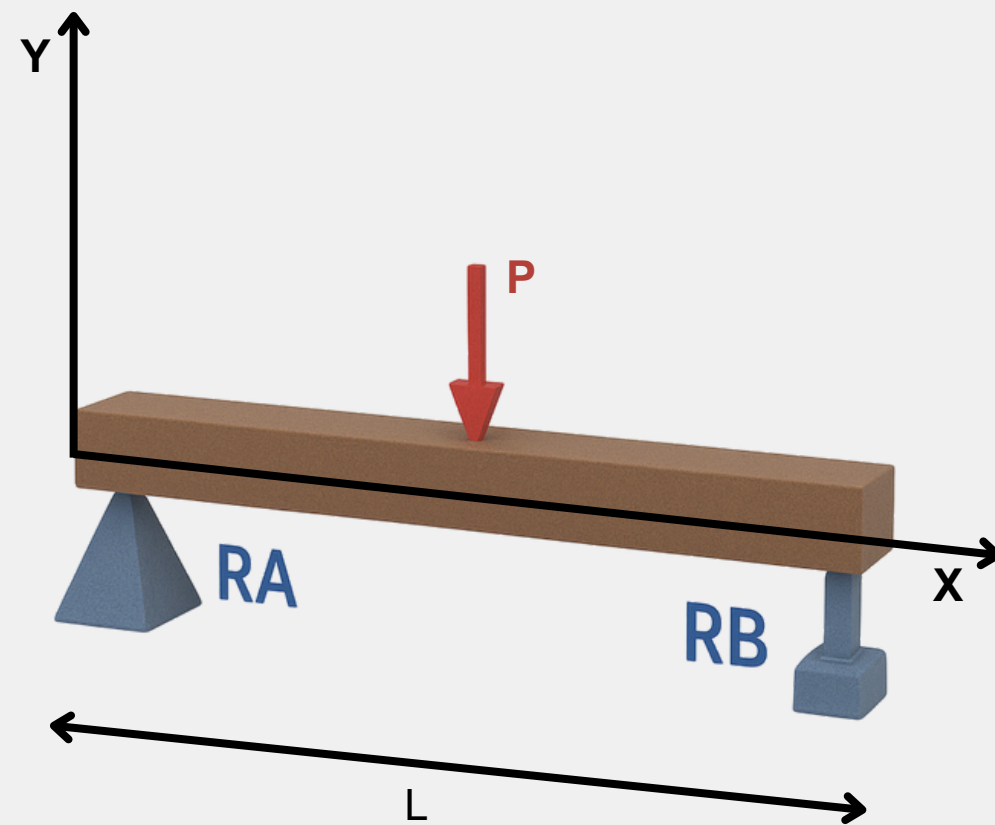
Key Points

- Even Distribution of Load: The load is spread uniformly along the entire length of the beam, leading to a consistent shear force and bending moment distribution.
- Deflection Shape: For simply supported beams, the deflection due to a UDL is parabolic, with the maximum deflection occurring at the midpoint of the beam.
- Shear and Bending Moment: The shear force varies linearly along the beam, while the bending moment increases quadratically toward the center (maximum moment at mid-span for simply supported beams).

Uniform Distributed Load



Point Load Mid-span



Equations

Reaction Forces
 $R_A = R_B = P/2$

Shear Force
For $0 \leq x \leq L/2$ $V(x) = P/2$
For $L/2 \leq x \leq L$ $V(x) = -P/2$

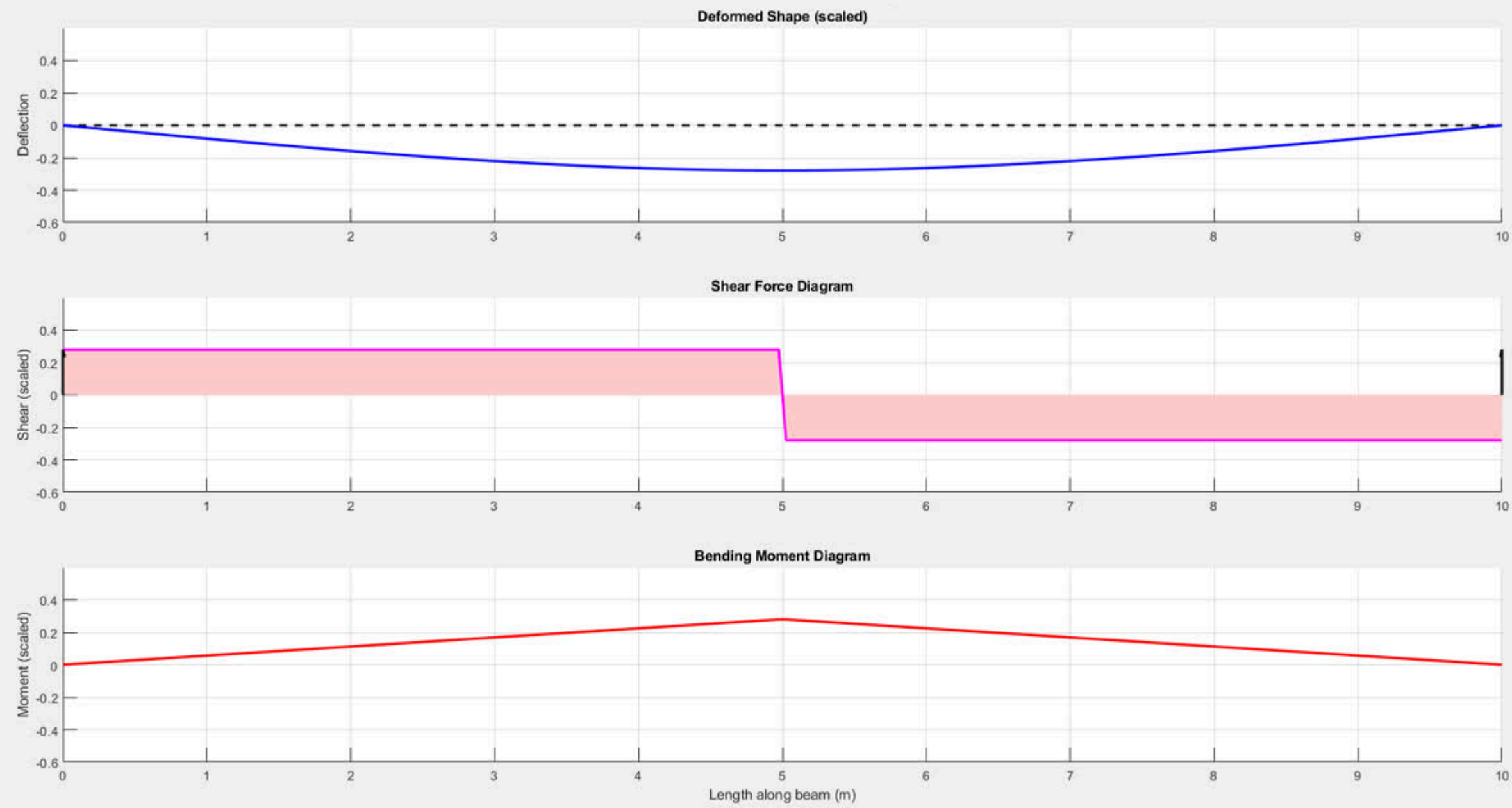
Bending Moment
For $0 \leq x \leq L/2$ $M(x) = Px/2$
For $L/2 \leq x \leq L$ $M(x) = P(L-x)/2$

Deflection
For $0 \leq x \leq L/2$ $\Delta y = (Px^3/12 - 3PL^2x/32)/EI$
For $L/2 \leq x \leq L$ $\Delta y = (PLx^2/4 - Px^3/12 - 5PL^2x/32 - 13PL^3/96)/EI$

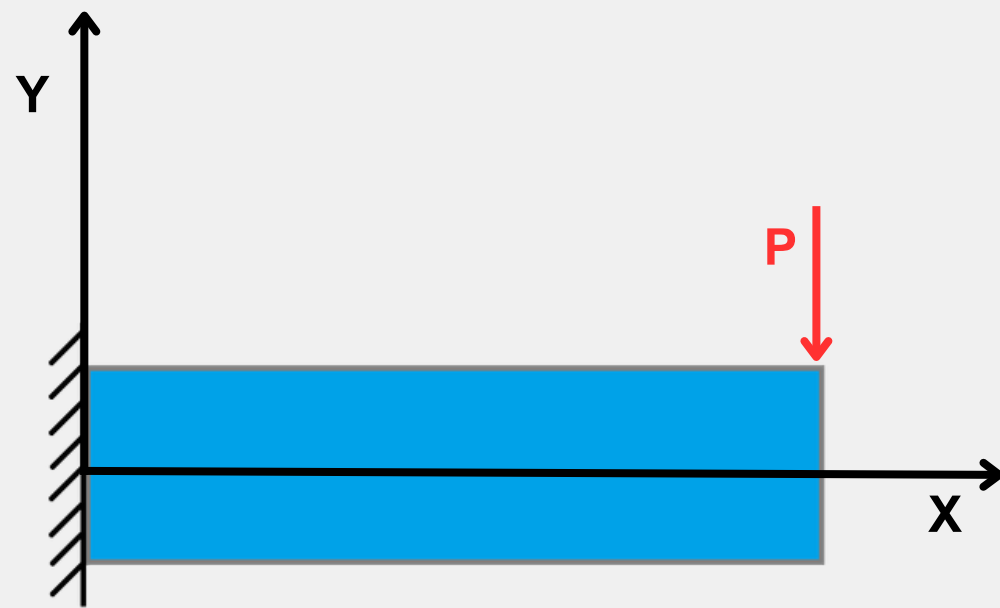
Key Points

- Concentration of Load: A point load is concentrated at a single point on the beam, typically at the mid-span in classical examples.
- Max Bending Moment: The maximum bending moment occurs at the point of application of the load and is equal to the load multiplied by half the span length (for simply supported beams).
- Deflection at Mid-Span: The deflection is greatest at the location of the load, which for mid-span point loads is the center of the beam.

Point Load Mid-span



Cantilever Beam Load



Equations

Shear Force $V(x)$:

$$V(x) = -p$$

Bending Moment $M(x)$:

$$M(x) = -p(l-x)$$

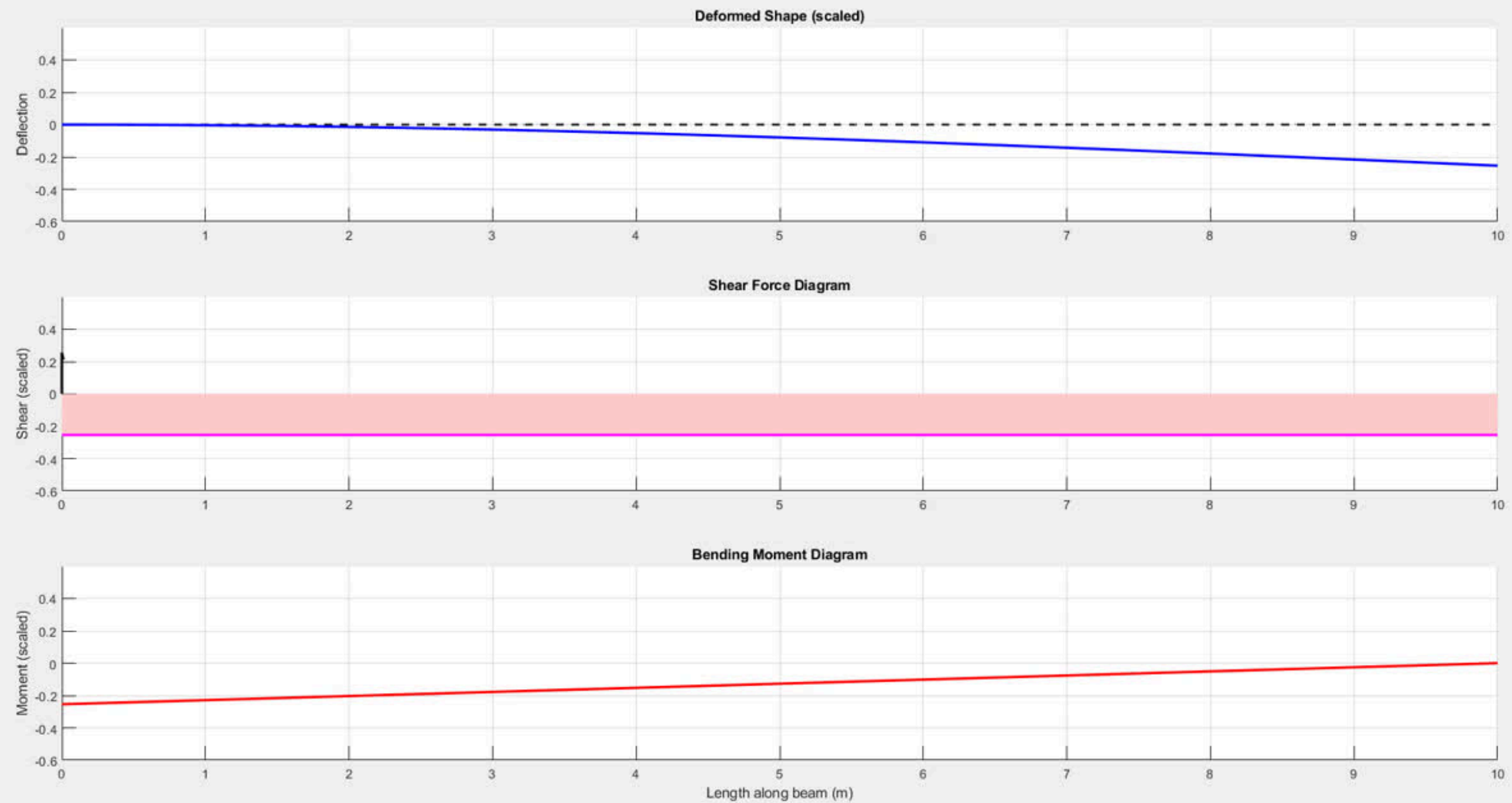
Deflection $y(x)$:

$$y(x) = -p/6EI (3Lx^2 - x^3)$$

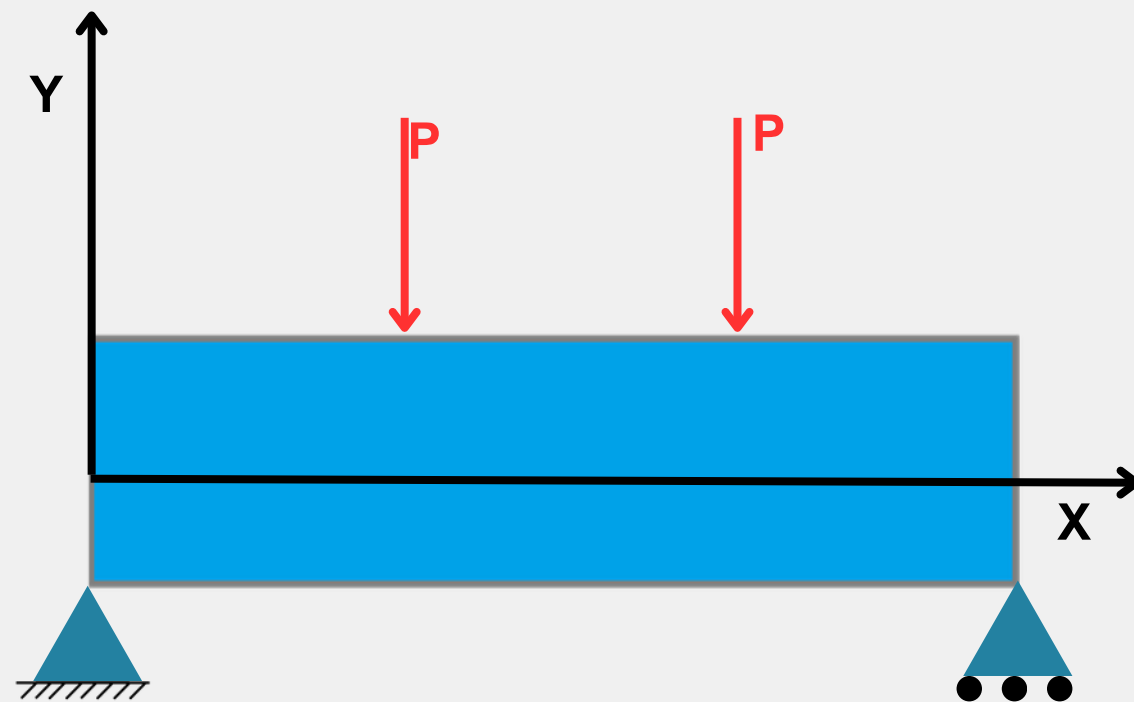
Key Points

- Max Moment at Fixed End: For a load applied at the free end of a cantilever beam, the maximum moment occurs at the fixed support, leading to a high stress concentration there.
- Deflection at Free End: The deflection is greatest at the free end of the cantilever, with a linear variation in deflection along the beam.
- Bending Moment Distribution: The bending moment is linear, with the moment being highest at the fixed end and decreasing to zero at the free end.

Cantilever Beam Load



Multiple Point Loads



Equations

Reaction Forces
 $R_A = R_B = P$

Shear Force
For $0 \leq x \leq L/3$ $V(x) = P$
For $L/3 \leq x \leq 2L/3$ $V(x) = 0$
For $2L/3 \leq x \leq L$ $V(x) = -P$

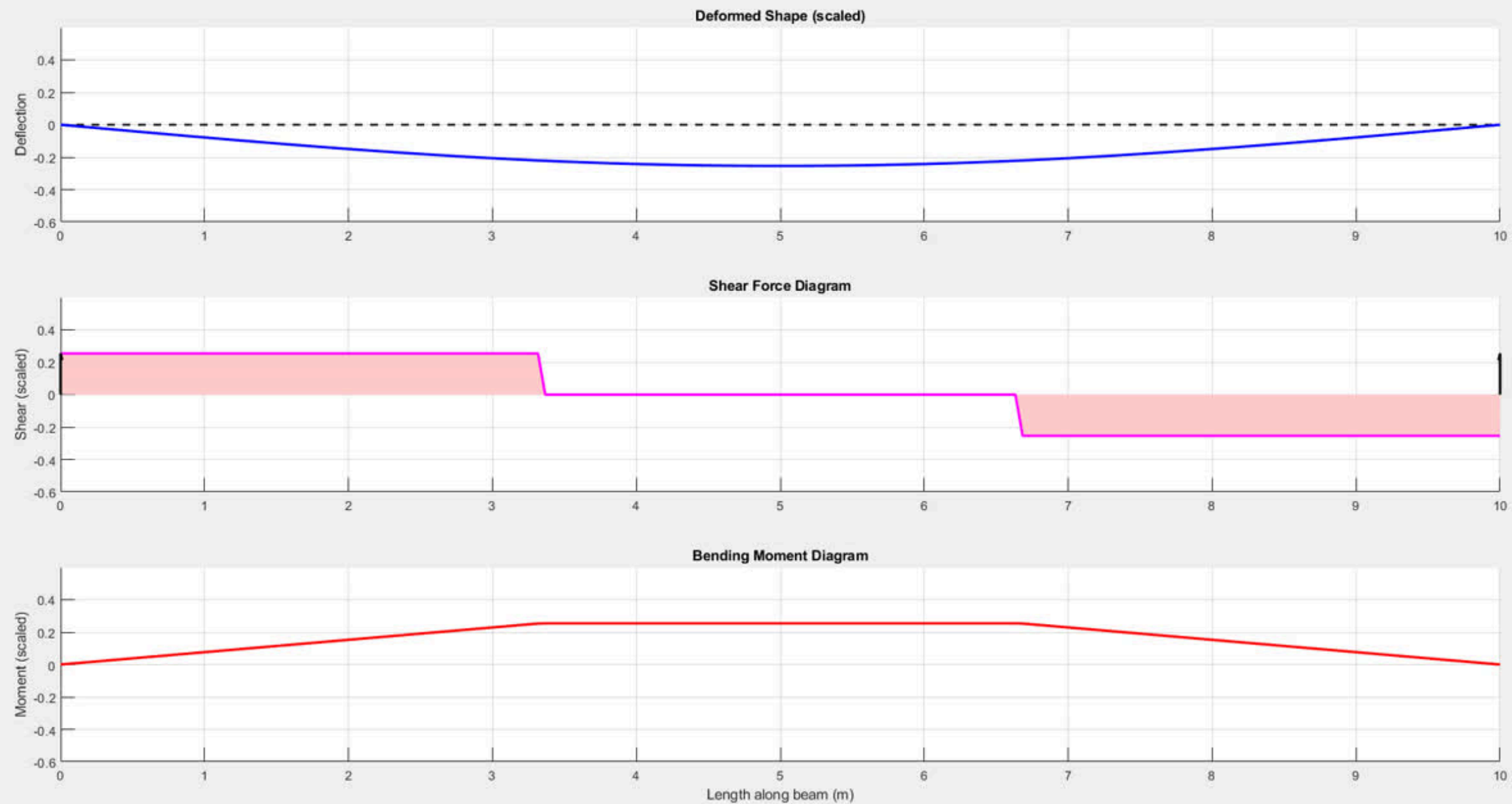
Bending Moment
For $0 \leq x \leq L/3$ $M(x) = Px$
For $L/3 \leq x \leq 2L/3$ $M(x) = PL/3$
For $2L/3 \leq x \leq L$ $M(x) = -P(x-L)$

Deflection
For $0 \leq x \leq L/3$ $y(x) = Px^3/6EI$
For $L/3 \leq x \leq 2L/3$ $y(x) = PLx^2/6EI - PL^2x/18EI + PL^3/162EI$
For $2L/3 \leq x \leq L$ $y(x) = -P(x-L)^3/6EI + 2L^2Px/9EI - PL^3/9EI$

Key Points

- Complex Shear Force Distribution: Point loads create discrete jumps in the shear force diagram at each point of load application.
- Variable Bending Moment: The bending moment diagram can have several peaks and valleys depending on the position of the point loads, and the moments will change abruptly at each load location.
- Interaction Between Loads: The overall deflection is the sum of the individual effects from each load, and the maximum deflection will typically occur between two loads or under the largest load.

Multiple Point Loads



Conclusion

- The project effectively fulfills its goal of visualizing shear force, bending moment and deflection for various beam loading condition using Matlab.
- We assessed how the deflection of beam, shear force and moment appears when load is applied on different types of beams.

The background is a light gray color, decorated with various hand-drawn blue doodles. These include several overlapping circles and loops at the top, a series of horizontal lines on the right side, a wavy line at the bottom center, and several checkmarks at the bottom right. There are also some scribbled shapes and a partial rainbow-like arc at the bottom left.

**Thank you
very much!**